

Case ID:

Case #001.

Title:

Windows Event Log Investigation.

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Date: 07 July 2026.

Status: Closed.

Severity: Informational.

1. Objective:

The objective of this project was to investigate Windows Security Event Logs using Event Viewer to understand successful authentication events and practice documenting security investigations in a professional manner.

2. Scenario:

During this investigation, Windows Event Viewer was used to review Security logs. Event ID 4624 (Successful Logon) was selected for analysis. The objective was to determine whether the event represented legitimate system activity or a potential security concern by examining authentication details and related event information.

3. Tools Used:

Tool	Purpose
Windows Event Viewer	Review and analyze Windows Security Events
Windows Security Log	Source of authentication events
Microsoft Windows 10	Operating system used during the investigation

4. Investigation Steps:

1. Opened Windows Event Viewer.
2. Navigated to **Windows Logs → Security**.
3. Filtered security events by **Event ID 4624**.
4. Identified a successful logon event.
5. Examined both the **General** and **Details** tabs.
6. Reviewed authentication information, process name, and workstation details.
7. Documented the findings and assessed the event.

5. Evidence:

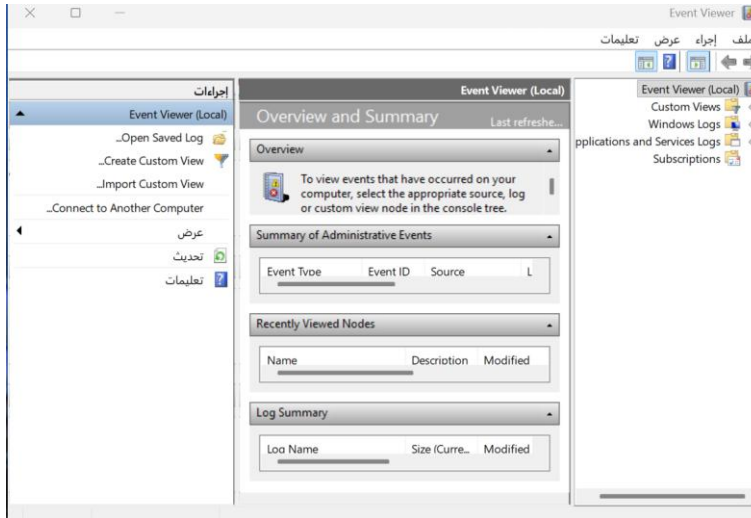


Figure 1. Windows Event Viewer interface.

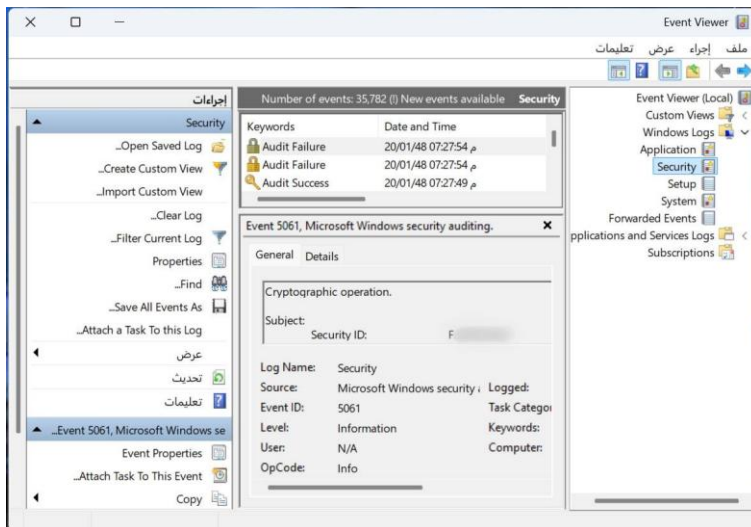


Figure 2. Windows Security Log containing recorded security events.

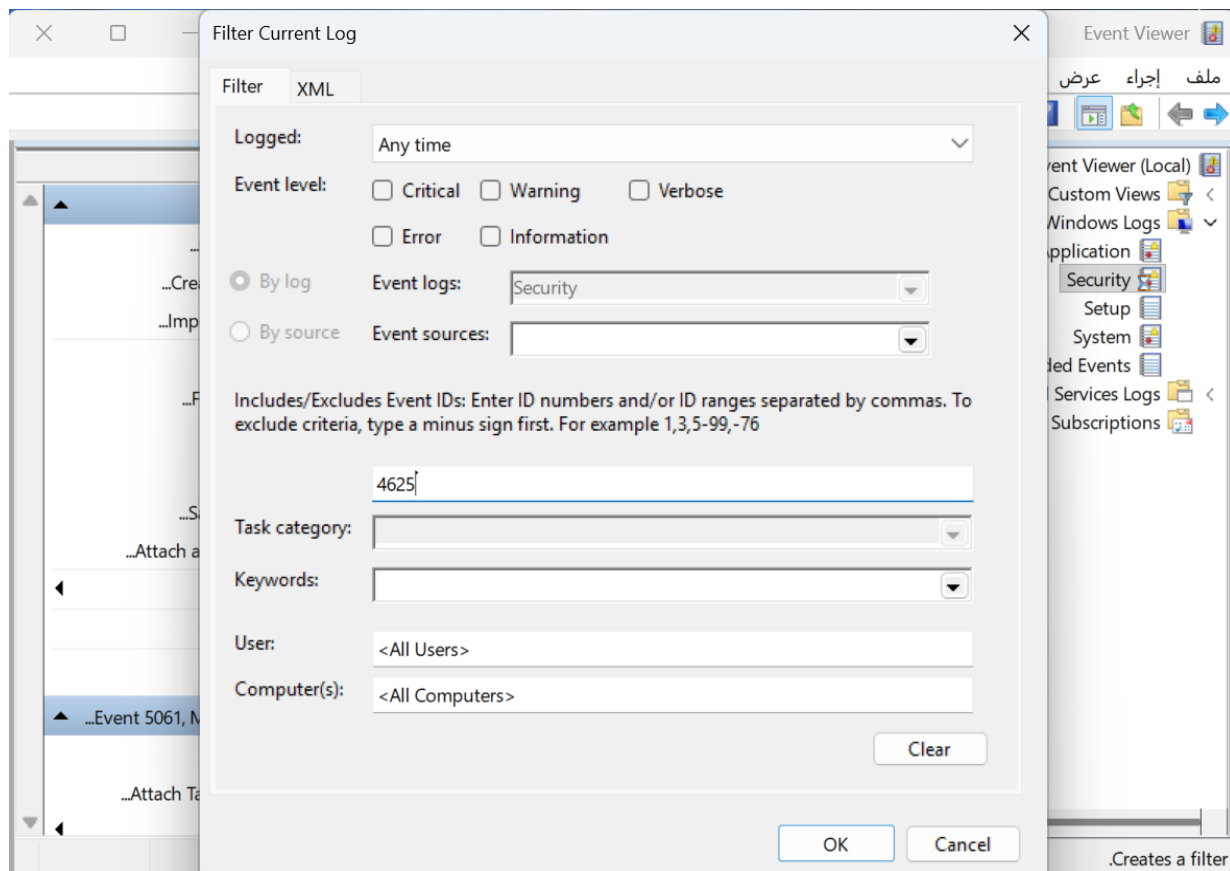


Figure 3. Filtered Security Log displaying Event ID 4624.

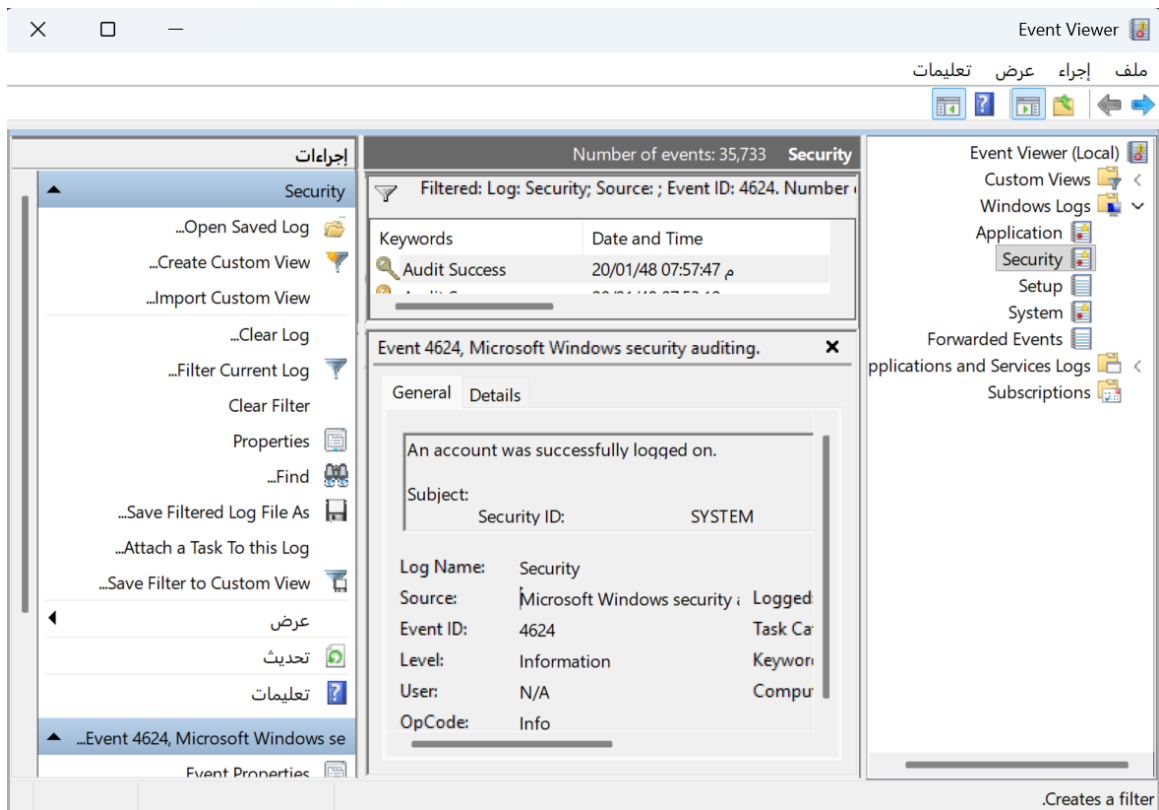


Figure 4. General view of Event ID 4624.

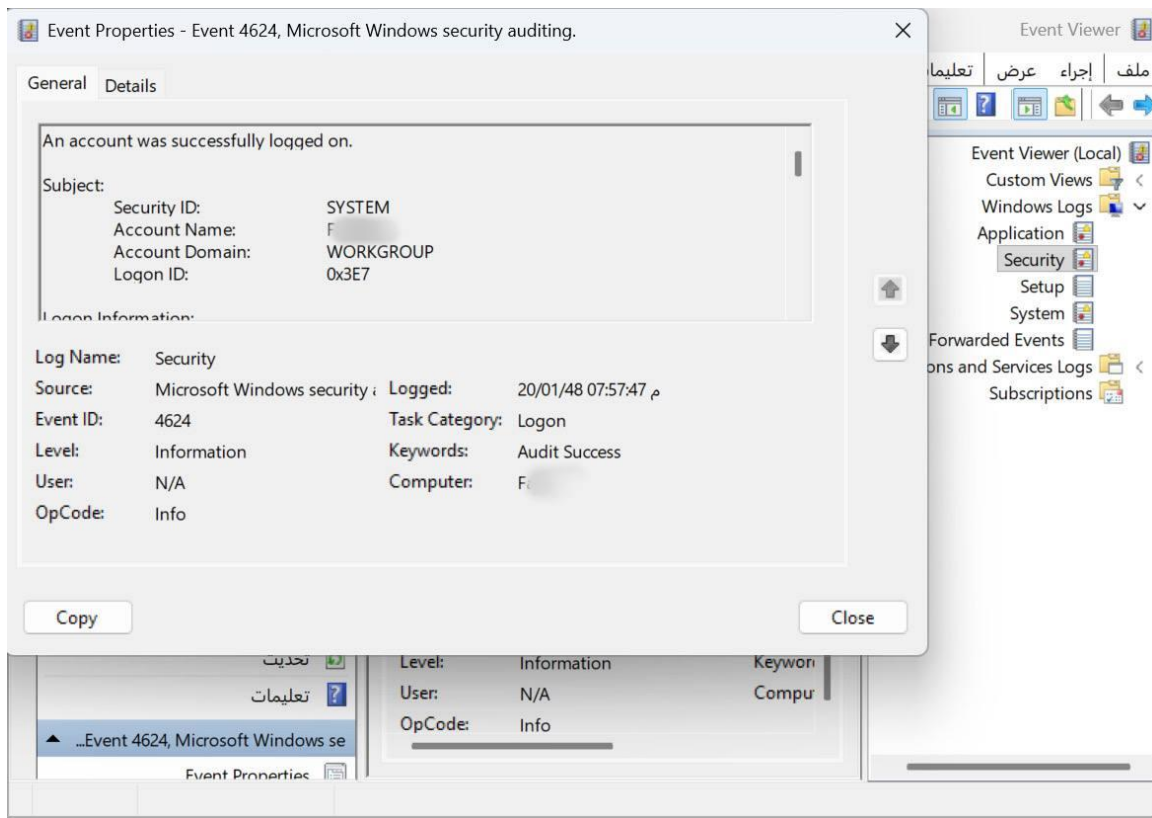


Figure 5. Detailed properties of Event ID 4624

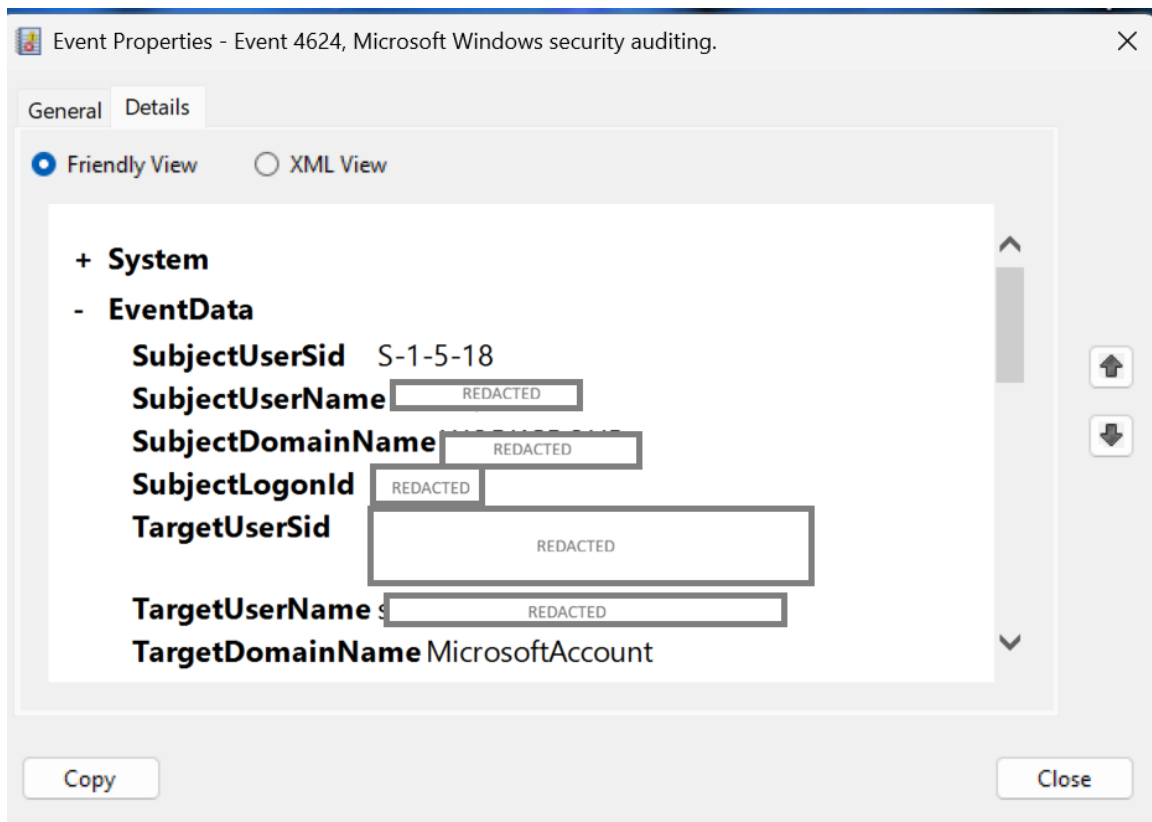


Figure 6. Authentication details for Event ID 4624.

6.Findings:

Item	Value
Event ID	4624
Event Type	Successful Logon
Logon Type	7 (Unlock Workstation)
Authentication Package	Negotiate
Logon Process	Negotiate
Process Name	C:\Windows\System32\lsass.exe
IP Address	Not Present
Assessment	Legitimate Activity

7. Analysis:

The investigation focused on Event ID 4624, which represents a successful logon event. The reviewed event was identified as a workstation unlock (Logon Type 7), indicating that an authorized user unlocked an existing session rather than initiating a new remote connection.

The authentication process was handled by the legitimate Windows security process (C:\Windows\System32\lsass.exe) using the Negotiate authentication package. No remote IP address, failed logon attempts, or other suspicious indicators were observed during the analysis.

Based on the available evidence, the event represents expected Windows operating system behavior and does not indicate malicious activity.

8. Conclusion:

This investigation successfully demonstrated how Windows Event Viewer can be used to analyze authentication events. The reviewed Event ID 4624 was determined to be a legitimate workstation unlock with no evidence of compromise or unauthorized access. The investigation highlights the importance of understanding Windows Security logs as part of routine security monitoring and incident analysis.

9. Recommendations:

Although no suspicious activity was identified during this investigation, the following security practices are recommended:

- Continue monitoring Windows Security logs for unusual authentication events.
- Regularly review failed logon events (Event ID 4625) to detect potential brute-force attempts.
- Keep Windows security updates installed to maintain system protection.
- Enable additional logging and auditing where appropriate to improve visibility during future investigations.
- Document security investigations consistently to support incident response and future analysis.

10. Skills Demonstrated:

- Windows Event Log Analysis.
- Event Viewer Investigation.
- Authentication Analysis.
- Security Log Review.
- Evidence Collection.
- Technical Documentation.
- Security Reporting.
- Analytical Thinking.

11. Key Takeaways:

1. Learned how to investigate Windows Security Event Logs.
2. Practiced identifying legitimate authentication events.
3. Improved technical documentation and reporting skills.